

Korff’s cylindric Hall–Littlewood weight is normalised at the wrong
end:

no cyclically ψ -graded weight carries Warnaar’s generalised
affine Jacobi–Trudi determinant

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Abstract

Warnaar [arXiv:2511.17034, §6] asks whether Huh–Kim–Krattenthaler–Okada’s combinatorial evaluation of a $t = 1$ affine bounded Littlewood determinant [arXiv:2301.13117, Thm. 3.3] extends to his generalised determinant (1) “by introducing an additional statistic on cylindric tableaux”. A previous note proved that no *monomial* statistic $t^{\text{stat}(T)}$ can do it, the obstruction being a cardinality count, and observed that the surviving possibility is a *polynomial* weight on each cylindric tableau, “exactly the shape of Macdonald’s classical $\psi_T(t)$ ”. Such a weight exists in the literature: Korff [arXiv:1110.6356, Def. 5.5] defines cylindric skew Hall–Littlewood functions $P_{\lambda/d/\mu}(x; t) = \sum_T \Psi_T(t) x^T$ with Ψ_T a product of factors $(1 - t^m)$ indexed by a *cyclic* ascent set. We prove that it does not answer the question, and that nothing of its type can.

The obstruction is the order of vanishing at $t = 1$. Because a non-constant cyclic 0/1 word necessarily contains a $0 \rightarrow 1$ ascent, Korff’s weight contributes a factor vanishing at $t = 1$ at *every* non-constant cylindric horizontal strip; a monomial prefactor $t^{\text{stat}(T)}$ is a unit at $t = 1$ and cannot repair this. At $k = 1$, $n = N = 2M$, $\alpha = (1^n)$ every cylindric tableau in the fibre has all n strips non-constant, forcing $\text{ord}_{t=1} c_\alpha \geq n = 2M$, whereas $c_\alpha(t) = C_M - (2M - 1)t^{M-1} + t^{M+1}$ has $\text{ord}_{t=1} c_\alpha = 2$ for $M = 2$ and 0 for every $M \geq 3$. So the identity fails for every $\ell \geq 0$, for every statistic and every choice of signs.

The failure is located exactly: at $\ell = 0$ the target *is* a ψ -weighted tableau sum — over the two *ordinary* semistandard tableaux of shape $(2, 2)$, each of whose ψ_T vanishes to order 2 at $t = 1$ — and passing to cylindric tableaux replaces Macdonald’s open ascent set by Korff’s cyclic one, which raises the order from 2 to 4. Equivalently, and this is the one-sentence summary: Korff’s deformation is normalised at $t = 0$ ($\Psi_T(0) = 1$, so $P_{\lambda/d/\mu}(x; 0)$ is the unweighted cylindric Schur function), while Warnaar’s needs the unweighted cylindric count at $t = 1$. We verify $\sum_T c^- \Psi_T(0) = c_\alpha(1)$ exactly, which is the identification of the two ends, and we record the consequence for the surviving type: an admissible weight must have $\text{ord}_{t=1} = 0$ on some tableau of each fibre, hence cannot be a product of factors $(1 - t^m)$ at all — it must be a ratio with as many factors above as below, a Gaussian-binomial type.

1 Holdings, and why this section exists

Every input below was read first-hand from a file on my own disk during this session. Recording it because the preceding six days of sessions repeatedly treated Korff [2] as something to be fetched while a 1,126,690-byte PDF with a text layer sat in `data/arxiv-rag/cylindric-partitions/`:

input	path
Korff 1110.6356 (77 pp)	data/arxiv-rag/cylindric-partitions/korff_1110.6356.pdf md5 f002ad7c129f1fff8a924afe7113eb7f
HKKO 2301.13117 (full L ^A T _E X source)	library/sources/20261009pm-cit/warnaar-cite_HKKO-2301.13117 md5 16d9ba13a138f5f92a076aa94dcfbae9
Warnaar 2511.17034 (full source)	library/sources/20261009pm-cit/warnaar-2511.17034-sigma26-0
Macdonald, 2nd ed. (486 pp)	library/macdonald-symmetric-functions-hall-polynomials.pdf

Korff’s title and abstract, `pdftotext -f 1 -l 1`, verbatim: *Cylindric versions of specialised Macdonald functions and a deformed Verlinde algebra*, arXiv:1110.6356v3 [math-ph], 6 Sep 2012. “We define cylindric generalisations of skew Macdonald functions $P_{\lambda/\mu}(q, t)$ when either $q = 0$ or $t = 0$. Fixing two integers $n > 2$ and $k > 0$ we shift the skew diagram λ/μ , viewed as a subset of the two-dimensional integer lattice, by the period vector $(n, -k)$.” The $q = 0$ branch is Hall–Littlewood: Korff’s (2.19)–(2.21) are Macdonald’s $\varphi_{\lambda/\mu}(t)$, $\psi_{\lambda/\mu}(t)$ verbatim, so this is the same specialisation as Warnaar’s, not an analogue of it.

2 The question, and what was already settled

Fix $k \geq 1$, $\ell \geq 0$, $n \geq 1$ and set

$$h := 2k, \quad w := 2\ell + 2, \quad M := k + \ell + 1, \quad N := h + w = 2M.$$

With $\dot{e}_r := e_r(x_1, x_1^{-1}, \dots, x_n, x_n^{-1})$, Warnaar [4, §6, (Eq-generalise)] studies

$$\mathcal{G}_{k,\ell,n}(x; t) := \sum_{y \in \mathbb{Z}^k} \det_{1 \leq i, j \leq k} \left(t^{My_i^2 - jy_i} (\dot{e}_{n-i+j-Ny_i}(x) - \dot{e}_{n+i+j-Ny_i}(x)) \right), \quad (1)$$

notes that at $t = 1$ this is the determinant evaluated combinatorially by Huh–Kim–Krattenthaler–Okada [1, Thm. 3.3] as

$$(x_1 \cdots x_n)^k \mathcal{G}_{k,\ell,n}(x; 1) = \sum_{\lambda \in \text{Par}(h, w)} c_{h,w}^-(\lambda) \sum_{T \in \text{CRST}(\lambda; h, w)} x^T, \quad (2)$$

and asks, verbatim: “**Is it possible to extend this result to (Eq-generalise) by introducing an additional statistic on cylindric tableaux?**” Here $\text{Par}(h, w) = \{\lambda : \ell(\lambda) \leq h, \lambda_1 - \lambda_h \leq w\}$, $\text{CRST}(\lambda; h, w)$ is the set of (h, w) -cylindric row-strict tableaux of shape λ [1, Defn. 2.1, Prop. 2.6], and

$$c_{h,w}^-(\lambda) = \begin{cases} 1, & \lambda_{2i-1} = \lambda_{2i} \text{ for all } 1 \leq i \leq k, \\ -1, & \lambda_1 - \lambda_h = w \text{ and } \lambda_{2i} = \lambda_{2i+1} \text{ for all } 1 \leq i \leq k-1, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Write, for a content $\alpha \in \mathbb{Z}_{\geq 0}^n$,

$$c_\alpha(t) := [x^\alpha] (x_1 \cdots x_n)^k \mathcal{G}_{k,\ell,n}(x; t).$$

What was settled. [5, Thm. 1] proves that no *monomial* statistic works: if $(x_1 \cdots x_n)^k \mathcal{G} = \sum_T \varepsilon(T) t^{\text{stat}(T)} x^T$ then $\|c_\alpha\|_1 \leq \#\{T : \text{cont } T = \alpha\}$, and at $k = 1$, $n = N = 2M$, $\alpha = (1^n)$ one has

$$c_\alpha(t) = C_M - (2M - 1)t^{M-1} + t^{M+1}, \quad A_\alpha = C_M, \quad B_\alpha = 2M - 2, \quad (4)$$

(C_M the M -th Catalan number, A_α and B_α the numbers of tableaux of content α with $c^- = +1, -1$), so that the coefficient $-(2M - 1)$ asks for one object more than exists. That note ends by observing that a *polynomial* weight $\pi_T(t) \in \mathbb{Z}[t]$ is not so obstructed, because $\|\sum_T \varepsilon(T) \pi_T\|_1$ is unbounded in terms of the number of tableaux once each π_T may have several terms of mixed sign — and that the natural candidate shape is Macdonald’s $\psi_T(t)$.

The capacity check, run before any modelling, and it passes. At $\ell = 0$ the target is $c_\alpha(t) = 2 - 3t + t^3$, $\|c_\alpha\|_1 = 6$, carried by 4 tableaux. A monomial weight gives $\|c_\alpha\|_1 \leq 4$: false, which is the whole of [5, Lem. 2]. A polynomial weight has no such bound — one factor $(1 - t^m)$ already has ℓ^1 -norm 2. *So the ℓ^1 capacity argument does not obstruct the polynomial type, and the test below is not pre-killed.* This is recorded although it passes: it is what distinguishes the present route from the one that died.

The question of this note. Korff [2, Def. 5.5] defines cylindric skew Hall–Littlewood functions as weighted sums over cylindric tableaux with a weight $\Psi_T(t)$ of exactly ψ -type. Is *that* the polynomial Warnaar’s §6 needs?

3 The dictionary

Korff’s parameters are not Warnaar’s, and one of the coincidences is dangerous. I write n_K, k_K for Korff’s two integers and reserve n for the number of variables.

Korff’s objects [2, §5.2]: $\lambda, \mu \in A_{k_K, n_K}^+ = \{(\lambda_1, \dots, \lambda_{k_K}) : n_K \geq \lambda_1 \geq \dots \geq \lambda_{k_K} \geq 1\}$, so k_K is a height and n_K a width bound; the period vector is $\Omega = (k_K, -n_K)$ in (row, column) coordinates, and a cylindric semistandard tableau satisfies $T(i, j) = T(i + k_K, j - n_K)$, weakly increasing along rows and strictly increasing down columns. His weight [2, (5.13)] is

$$\Psi_{\lambda/d/\mu}(t) = \prod_{i \in J_{\lambda/d/\mu}} (1 - t^{m_i(\mu)}), \quad J_{\lambda/d/\mu} = \{i : \theta_i = 0, \theta_{i+1} = 1\}, \quad (5)$$

$\theta_i = \lambda'[d]_i - \mu'[0]_i$, $m_i(\mu) = \mu'_i - \mu'_{i+1}$, and the index set is **cyclic of period n_K** : Korff’s own proof of his Lemma 5.4 uses $\theta_{n_K+1} = \lambda'[d]_1 - \mu'[0]_1$, and $\sum_{i \in \mathbb{Z}/n_K\mathbb{Z}} m_i(\lambda) = k_K$, i.e. $m(\lambda)$ is a configuration of k_K particles on n_K cyclic sites. Ψ_T is the product of (5) over the decomposition of T into cylindric horizontal strips.

HKKO’s lattice-path model [1, Prop. 2.7]: for $\lambda \in \text{Par}(h, w)$, $\#\{T \in \text{CRST}(\lambda; h, w) : \text{cont } T = \alpha\}$ is the number of paths in $\mathbb{Z}^h$ from 0 to $(\lambda_1, \dots, \lambda_h)$ staying in

$$R := \{x \in \mathbb{Z}^h : x_1 \geq x_2 \geq \dots \geq x_h \geq x_1 - w\},$$

the a -th step being a 0/1 vector with α_a ones. The $x^{(a)}$ are the row-length vectors of the subtableaux of entries $\leq a$.

Proposition 1 (dictionary). $n_K = h = 2k$ and $k_K = w = 2\ell + 2$. Explicitly, for $x \in R$ put

$$m_i(x) = x_i - x_{i+1} \quad (1 \leq i < h), \quad m_h(x) = w - (x_1 - x_h); \quad (6)$$

then $m(x) \in \mathbb{Z}_{\geq 0}^h$ with $\sum_{i=1}^h m_i(x) = w$, and a step $x \mapsto x + \theta$, $\theta \in \{0, 1\}^h$, satisfies

$$m_i(x + \theta) - m_i(x) = \theta_i - \theta_{i+1} \quad (i \in \mathbb{Z}/h\mathbb{Z}, \theta_{h+1} := \theta_1). \quad (7)$$

Korff's column index i is the row index of the HKKO row-strict tableau, and $m_i(\mu) = x_i - x_{i+1}$.

Proof. Two independent routes agree.

(i) *Shape parameters.* By [1, Prop. 2.4], transposition is a bijection $\text{CRST}(\lambda; h, w) \rightarrow \text{CSSYT}(\text{Tr}(\lambda; h, w); w, h)$; so in the semistandard picture — Korff's — the period is $(w, -h)$, i.e. height w and width h . Matching against Korff's period $(k_K, -n_K)$ gives $k_K = w$, $n_K = h$.

(ii) *State count.* For $x \in R$, (6) gives $m_i \geq 0$ by $x_i \geq x_{i+1}$ for $i < h$ and by $x_h \geq x_1 - w$ for $i = h$; and $\sum_i m_i = (x_1 - x_h) + w - (x_1 - x_h) = w$. So R maps onto configurations of w particles on h cyclic sites, which is Korff's level- k_K weight lattice on n_K sites with $k_K = w$, $n_K = h$. Identity (7) is immediate from (6) for $i < h$, and for $i = h$ reads $m_h(x + \theta) - m_h(x) = -(\theta_1 - \theta_h) = \theta_h - \theta_{h+1}$ with $\theta_{h+1} = \theta_1$. This is verbatim Korff's relation $\theta_i - \theta_{i+1} = m_i(\lambda) - m_i(\mu)$ in the proof of his Lemma 5.4.

Finally, in the semistandard picture θ'_i is the number of boxes added in column i , which is the number added in row i of the row-strict tableau, i.e. θ_i ; and μ'_i , the i -th column length of the semistandard tableau, is the i -th row length x_i of the row-strict one, so $m_i(\mu) = \mu'_i - \mu'_{i+1} = x_i - x_{i+1}$. $\square$

So Korff's weight transported to HKKO's tableaux is: for a path $P = (x^{(0)} = 0, x^{(1)}, \dots, x^{(n)})$ in R with steps $\theta^{(a)} = x^{(a)} - x^{(a-1)} \in \{0, 1\}^h$,

$$\Psi_P(t) := \prod_{a=1}^n \prod_{i \in J(\theta^{(a)})} (1 - t^{m_i(x^{(a-1)})}), \quad J(\theta) = \{i \in \mathbb{Z}/h\mathbb{Z} : \theta_i = 0, \theta_{i+1} = 1\}. \quad (8)$$

Write $\mathcal{T}_\alpha$ for the set of such paths of content α and $\lambda(P) = x^{(n)}$.

Lemma 2. If $i \in J(\theta^{(a)})$ then $m_i(x^{(a-1)}) \geq 1$, so (8) is a product of factors $(1 - t^m)$ with $m \geq 1$. In particular $\Psi_P(0) = 1$ for every P .

Proof. Write $x = x^{(a-1)}$, $\theta = \theta^{(a)}$, $y = x + \theta \in R$. For $i < h$, $\theta_i = 0$ and $\theta_{i+1} = 1$ give $y_i = x_i$, $y_{i+1} = x_{i+1} + 1$, and $y_i \geq y_{i+1}$ forces $m_i(x) = x_i - x_{i+1} \geq 1$. For $i = h$, $\theta_h = 0$ and $\theta_1 = 1$ give $y_h = x_h$, $y_1 = x_1 + 1$, and $y_h \geq y_1 - w$ forces $m_h(x) = w - (x_1 - x_h) \geq 1$. $\square$

Validation of (8), with controls

Four readings of (5) were implemented: the one above (`true_Psi`); `open_J`, with the non-cyclic index set $\{1 \leq i < h\}$, which is Macdonald's; `shift_m`, using m_{i+1} in place of m_i ; `use_mlam`, using the outer shape's multiplicities; and `I_with_mu`, using Korff's I -set (5) with $m(\mu)$. Two gates, each with the controls run:

reading	violations of Korff (5.16) (180 strips, $h \leq 4, w \leq 3$)	departure order from Macdonald's ψ ($h = 2, w = 8$ / $h = 3, w = 9$)
<code>true_Psi</code>	0	6 / 7
<code>open_J</code>	36	— (equals Macdonald by definition)
<code>shift_m</code>	88	0 / 0
<code>use_mlam</code>	114	0 / 0
<code>I_with_mu</code>	88	0 / 0

Gate 1 is Korff's printed identity [2, (5.16)], $b_\lambda \Psi_{\lambda/d/\mu} = b_\mu \Phi_{\lambda/d/\mu}$ with $b_\lambda = \prod_i (t)_{m_i(\lambda)}$; it is passed by the reading above and failed by all four variants, *including* the non-cyclic one, so the cyclicity of the index set is pinned by Korff's own lemma and not by my guess. Gate 2 is the non-cylindric reduction: the correct weight differs from Macdonald's ordinary ψ_T only through the single wrap factor $(1 - t^{m_h})$, $m_h = w - (x_1 - x_h)$, so it must depart from ψ_T only at order $\geq w - \text{span}$ — measured 6 and 7 against $w = 8, 9$ — whereas a misplaced wrap departs at order 0.

Remark 3 (a gate that turned out vacuous, recorded as such). I first used the symmetry of $\sum_{P:\lambda(P)=\lambda} \Psi_P x^{\text{cont } P}$ as the gate: Korff's $P_{\lambda/d/\mu}$ is a symmetric function, and a wrong index set should break it. Measured: symmetric for all 127 cylindric shapes tested at $(h, w, n) \in \{(2, 2, 3), (2, 4, 3), (3, 2, 3), (4, 2, 3), (3, 3, 3), (2, 2, 4), (4, 2, 4)\}$ — *and also symmetric for every variant*, including `open_J` and Φ . So the symmetry gate is a constant function of the question asked and reports nothing; the tables above replace it.

Remark 4 (Korff's Example 5.1 is not usable as a gate). Korff's Example 5.1 ($n_K = 5, k_K = 6$, three strips) lists index sets $\{3, 6\}, \{4\}, \{4, 6\}$, five elements in all, and prints the resulting weight as $(1 - t)^4(1 - t^2)^2$, which is a product of *six* factors. Five index elements cannot give six factors. Either an element was dropped (most likely $\{4\}$ should read $\{4, 6\}$, which would give $2 + 2 + 2 = 6$) or the printed weight is for a different decomposition; and the listed sets contain $6 > n_K = 5$, which the stated index range $1 \leq i \leq n_K$ excludes. I cannot decode it from the text I hold and therefore did not use it. This is the only datum in the paper I wanted and could not obtain.

Remark 5 (numbering audit, and a self-inflicted scare). Every HKKO number cited here was recounted from the source this session rather than recalled. HKKO declare `\newtheorem{thm}{Theorem}[section]` with every other environment sharing that counter, so numbers are determined by the order of `\begin{...}` lines. Counted: `defn:cylindric` is Defn. 2.1, `def:par` Defn. 2.2, `prop:cssyt=crst` Prop. 2.4, `defn:cSchur` Defn. 2.5, `prop:cyl_sch` Prop. 2.6, `prop:cssyt=path` Prop. 2.7, `thm:ssyt2` Thm. 2.8, and `thm:C` — the theorem Warnaar cites — Thm. 3.3.

My first count made `thm:C` Theorem 3.2, which would have meant Warnaar's "[HKKO25, Theorem 3.3]" named a different theorem and that the whole $t = 1$ gate of [5] rested on the wrong result. The error was mine: my grep pattern listed `thm, cor, conj, lem, assumption, prop, defn, quest, problem, rem` and omitted `rems`, and there is a `\begin{rems}` at source line 1113 which consumes counter 3.2. *An enumeration of an environment list is an instrument, and a missing alternative in it shifts every subsequent number by one.* Recorded because the first reading was alarming and wrong, and because two numbers in [5] are wrong in the other direction: it cites Prop. 2.9 for `prop:cyl_sch` (it is 2.6) and Thm. 2.10 for `thm:ssyt2` (it is 2.8). Its Prop. 2.7 for the path model is right. No statement in either paper depends on a number: every input was quoted and, in the case of eq. (3.6), verified symbolically at six parameter points against three deliberately wrong variants.

Remark 6 (two of Korff's own standing restrictions exclude the target). Korff assumes $n_K > 2$ throughout, which by Proposition 1 is $2k > 2$, i.e. $k \geq 2$; the witness family of (4) has $k = 1$, so it sits at $n_K = 2$. And Korff's Def. 5.5 defines $P_{\lambda/d/\mu}(x_1, \dots, x_{\ell_K}; t)$ for $1 \leq \ell_K \leq k_K$ and sets it to zero for $\ell_K > k_K$; here $\ell_K = n$ is the number of variables and $k_K = w$, and the witness has $n = N = 2k + w > w$, so Korff's own convention makes his function vanish identically at every parameter where Warnaar's question has content. I drop that convention — formula (8) is defined for any number of variables — and test the *formula*, because answering a mathematical question by a convention would be no answer. The combinatorics is visibly non-empty there: e.g. $h = w = 2$, $n = 4$ has 4 cylindric tableaux of content (1^4) . These two restrictions are recorded, not used.

4 The obstruction: order of vanishing at $t = 1$

Definition 7. Let $\nu(P) := \sum_{a=1}^n |J(\theta^{(a)})|$. A weight assignment $\pi : \mathcal{T} \rightarrow \mathbb{Z}[t, t^{-1}]$ is *cyclically ψ -graded* if for every P either $\pi_P = 0$ or $\text{ord}_{t=1} \pi_P \geq \nu(P)$, where $\text{ord}_{t=1} f$ denotes the multiplicity of $t = 1$ as a root of f .

Lemma 8. For every P and every $s(P) \in \mathbb{Z}$, $\varepsilon(P) \in \mathbb{Z} \setminus \{0\}$, the weight $\pi_P = \varepsilon(P) t^{s(P)} \Psi_P(t)$ is cyclically ψ -graded, with $\text{ord}_{t=1} \pi_P = \nu(P)$ exactly.

Proof. By Lemma 2, Ψ_P is a product of exactly $\nu(P)$ factors $(1 - t^m)$ with $m \geq 1$, each vanishing to order exactly 1 at $t = 1$; and $\varepsilon(P) t^{s(P)}$ is a unit at $t = 1$. $\square$

The next lemma is the whole mechanism, and the only place cyclicity is used.

Lemma 9 (cyclic ascent). For $\theta \in \{0, 1\}^h$ one has $J(\theta) = \emptyset$ if and only if θ is constant, i.e. $\theta = (0, \dots, 0)$ or $\theta = (1, \dots, 1)$. Hence $\nu(P) \geq \#\{a : \theta^{(a)} \text{ non-constant}\} = \#\{a : \alpha_a \notin \{0, h\}\}$.

Proof. If θ is constant there is no i with $\theta_i = 0, \theta_{i+1} = 1$. Conversely let θ be non-constant, so the sets $Z = \{i : \theta_i = 0\}$ and $O = \{i : \theta_i = 1\}$ are both non-empty. Pick $p \in O$. Walking backwards from p around the cycle $\mathbb{Z}/h\mathbb{Z}$ one must leave O , since $Z \neq \emptyset$; let i be the first index met with $\theta_i = 0$. Then $\theta_{i+1} = 1$, so $i \in J(\theta)$. The last equality holds because $\sum_i \theta_i = \alpha_a$, and a 0/1 vector of length h is constant exactly when its sum is 0 or h . $\square$

Lemma 10. Let $h \geq 2$ and $\alpha = (1^n)$. Then $\nu(P) = n$ for every $P \in \mathcal{T}_\alpha$.

Proof. Each step has $\sum_i \theta_i^{(a)} = 1$, so $\theta^{(a)}$ has a single 1, at some position p_a ; since $h \geq 2$ it is non-constant. The only $i \in \mathbb{Z}/h\mathbb{Z}$ with $\theta_i^{(a)} = 0$ and $\theta_{i+1}^{(a)} = 1$ is $i = p_a - 1$, so $|J(\theta^{(a)})| = 1$ for each of the n steps. $\square$

Theorem 11. Let $k \geq 1$, $\ell \geq 0$, $h = 2k$, $w = 2\ell + 2$, $n \geq 1$, and let $\pi : \mathcal{T} \rightarrow \mathbb{Z}[t, t^{-1}]$ be cyclically ψ -graded. Suppose

$$(x_1 \cdots x_n)^k \mathcal{G}_{k, \ell, n}(x; t) = \sum_{P \in \mathcal{T}} \pi_P(t) x^{\text{cont } P}. \quad (9)$$

Then for every content α with $c_\alpha \neq 0$,

$$\text{ord}_{t=1} c_\alpha(t) \geq \min\{\nu(P) : P \in \mathcal{T}_\alpha, \pi_P \neq 0\}.$$

Proof. Comparing the coefficient of x^α in (9) gives $c_\alpha = \sum_{P \in \mathcal{T}_\alpha} \pi_P$. Each summand is either 0 or divisible by $(t-1)^{\nu(P)}$ in $\mathbb{Z}[t, t^{-1}]$, hence the sum is divisible by $(t-1)^{\min \nu(P)}$, the minimum over those P with $\pi_P \neq 0$. As $c_\alpha \neq 0$ its order of vanishing is finite and at least that exponent. $\square$

Corollary 12 (Korff’s weight fails, uniformly in ℓ). *Let $k = 1$, $\ell \geq 0$, $M = \ell + 2$, $n = N = 2M$, $\alpha = (1^n)$. There is no $\varepsilon : \mathcal{T} \rightarrow \mathbb{Z}$ and no $\text{stat} : \mathcal{T} \rightarrow \mathbb{Z}$ with*

$$(x_1 \cdots x_n) \mathcal{G}_{1,\ell,n}(x; t) = \sum_{P \in \mathcal{T}} \varepsilon(P) t^{\text{stat}(P)} \Psi_P(t) x^{\text{cont } P}.$$

In particular Huh–Kim–Krattenthaler–Okada’s Theorem 3.3 with the cylindric Schur function replaced by Korff’s cylindric Hall–Littlewood function — with or without an additional monomial statistic, with HKKO’s signs (3) or any others — is false for every $\ell \geq 0$.

Proof. Suppose such ε, stat exist. By Lemma 8 the weight is cyclically ψ -graded, and $\mathcal{T}_\alpha \neq \emptyset$ (it contains the path with $\theta^{(a)} = e_1$ for $a \leq M$ and e_2 for $a > M$, whose endpoint is (M, M)). By (4), $c_\alpha \neq 0$. By Lemma 10, $\nu(P) = n = 2M$ for every $P \in \mathcal{T}_\alpha$, since $h = 2k = 2 \geq 2$. So Theorem 11 forces

$$\text{ord}_{t=1}(C_M - (2M - 1)t^{M-1} + t^{M+1}) \geq 2M. \quad (10)$$

Evaluate the left side. At $t = 1$ the polynomial takes the value $C_M - 2M + 2$. For $M = 2$ this is $2 - 4 + 2 = 0$, and indeed $2 - 3t + t^3 = (1 - t)^2(t + 2)$, whose order of vanishing at $t = 1$ is exactly 2 because $(t + 2)|_{t=1} = 3 \neq 0$; and $2 < 4 = 2M$. For $M \geq 3$ we claim $C_M > 2M - 2$, so that the order is $0 < 2M$. Indeed $C_3 = 5 > 4$, and $C_{M+1}/C_M = 2(2M + 1)/(M + 2) \geq 2$ for $M \geq 1$, so $C_M \geq 5 \cdot 2^{M-3}$ for $M \geq 3$, while $2M - 2 < 5 \cdot 2^{M-3}$ for all $M \geq 3$ (at $M = 3$: $4 < 5$; and the right side at least doubles while the left grows by 2). In either case (10) fails. $\square$

Remark 13 (the same argument at $k \geq 2$). Lemmas 9 and 10 are uniform in k , so Theorem 11 applies verbatim; only the closed form (4) is special to $k = 1$. Computed values of c_α at $\alpha = (1^n)$ from (1): $(k, \ell, n) = (2, 0, 4)$ gives $3t^2 - 6t + 3 = 3(t - 1)^2$, order 2, against the required 4; $(2, 1, 6)$ gives $5t^3 - 15t^2 + 14$, order 0, against the required 6. Two further rows are **vacuous and are recorded as such**: at $(2, 0, 5)$ and $(2, 1, 7)$ one has $c_\alpha \equiv 0$, so Theorem 11 says nothing there. Honest count: of six (k, ℓ, n) tested with $k \geq 2$, four are informative and two are empty.

5 Where the failure lives, exactly

The obstruction is not that the target dislikes ψ -weights. At $\ell = 0$ it is a ψ -weighted tableau sum — over *ordinary* tableaux.

Proposition 14. *At $k = 1$, $\ell = 0$, $n = 4$, $\alpha = (1^4)$:*

1. *Warnaar’s Conjecture 1.5 asserts $(x_1 \cdots x_4) \mathcal{G}_{1,0,4}(x; t) = \sum_{\lambda \text{ even}, \lambda_1 \leq 2} P_\lambda(x; t)$, whose x^α -coefficient is $\sum_T \psi_T(t)$ over the two ordinary semistandard tableaux of shape $(2, 2)$ and content (1^4) . Their Macdonald weights are $(1 - t)^2$ and $(1 - t)^2(1 + t)$; each vanishes to order exactly 2 at $t = 1$, i.e. at exactly 2 of its 4 horizontal strips; and their sum is $(1 - t)^2(t + 2) = 2 - 3t + t^3 = c_\alpha(t)$.*
2. *The four $(2, 2)$ -cylindric tableaux of the same content all carry the same Korff weight $\Psi_P = (1 - t)^2(1 - t^2)^2$, of order 4 at $t = 1$: the cyclic ascent set fires at every one of the 4 strips.*

3. Consequently $\sum_P c^-(\lambda(P))\Psi_P = 0 \neq c_\alpha(t)$, and no statistic repairs it: $c_\alpha/\Psi = (t+2)/((1-t)^2(1+t)^2)$ is not a Laurent polynomial.

So the entire discrepancy is the integer 2 versus 4: passing from Macdonald's open ascent set $\{1 \leq i < h : \theta_i = 0, \theta_{i+1} = 1\}$ to Korff's cyclic one $\{i \in \mathbb{Z}/h\mathbb{Z} : \dots\}$ adds the wrap index $i = h$, and for a strip of size 1 in a 2-row cylinder the wrap index fires exactly when the open one does not. Each of the 4 strips therefore contributes a vanishing factor instead of 2 of them doing so.

Proposition 15 (the two deformations are normalised at opposite ends). $\Psi_P(0) = 1$ for every P (Lemma 2), so Korff's $P_{\lambda/d/\mu}(x; 0)$ is the unweighted cylindric Schur function; whereas (2) says that it is at $t = 1$ that Warnaar's determinant becomes the unweighted cylindric count. Accordingly $\sum_{P \in \mathcal{T}_\alpha} c^-(\lambda(P))\Psi_P(0) = A_\alpha - B_\alpha = c_\alpha(1)$ for every α : verified at $(k, \ell, n) = (1, 0, 4)$ and $(1, 1, 6)$, where the two sides are $0 = 0$ and $1 = 1$.

Corollary 16 (what a correct weight must look like). Let π be any weight satisfying (9). At $k = 1$, $\alpha = (1^{2M})$ there must exist $P \in \mathcal{T}_\alpha$ with $\text{ord}_{t=1} \pi_P \leq 2$ ($M = 2$) and with $\text{ord}_{t=1} \pi_P = 0$ ($M \geq 3$). So for $M \geq 3$ at least one cylindric tableau of the fibre carries a weight that does not vanish at $t = 1$, hence is not a product of factors $(1 - t^m)$, $m \geq 1$, at all. A weight of the surviving type must be a ratio with as many such factors above as below — a Gaussian-binomial type, whose value at $t = 1$ is the corresponding ordinary binomial — so that the $t = 1$ specialisation returns HKKO's unweighted count. This is a genuine narrowing: it rules out ψ_T and φ_T and Korff's Ψ_T, Φ_T together with every monomial regrading of them, and it points at the modified Hall–Littlewood / Kostka–Foulkes normalisation rather than at the branching coefficients.

6 Verification

All computations are exact (SymPy, rational arithmetic); code in `proofs/code-1010-q410/`.

1. **Ground truth re-derived.** c_α at $\alpha = (1^n)$ computed directly from (1) by the determinant engine agrees with (4) at $(k, \ell, n) = (1, 0, 4)$ and $(1, 1, 6)$: $t^3 - 3t + 2$ and $t^4 - 5t^2 + 5$, with $\text{ord}_{t=1} = 2$ and 0.
2. $\nu(P) = n$ **at** $\alpha = (1^n)$ (Lemma 10) verified by enumeration: the set of values of ν over all $P \in \mathcal{T}_{(1^n)}$ is the singleton $\{n\}$ at $(k, \ell, n) = (1, 0, 4), (1, 1, 6), (1, 2, 8), (2, 0, 4), (2, 1, 6), (3, 0, 6)$, over 4, 18, 68, 6, 64, 20 tableaux respectively.
3. **The $t = 1$ evaluation, brute force.** Independently of Lemma 9, for each content α the signed count $\sum \{c^-(\lambda(P)) : \Psi_P(1) \neq 0\}$ was computed by evaluating (8) at $t = 1$ on every tableau, and compared with the prediction “+1 if every $\alpha_a \in \{0, h\}$, else 0” that Lemma 9 gives. Agreement in **23891 of 23891** contents at $(k, \ell, n) = (1, 0, 4), (1, 1, 6), (1, 0, 3), (1, 1, 5), (2, 0, 4), (2, 1, 6)$
4. **The violations.** Comparing that signed count with $c_\alpha(1) = A_\alpha - B_\alpha$:

(k, ℓ, n)	contents	contents violating	first violating α
(1, 0, 3)	27	0	—
(1, 0, 4)	81	0	—
(1, 1, 5)	243	90	$(0, 0, 0, 1, 1), c_\alpha(1) = 1$ vs 0
(1, 1, 6)	729	301	$(0, 0, 0, 0, 1, 1), c_\alpha(1) = 1$ vs 0
(1, 2, 8)	6561	3025	$(0^6, 1, 1), c_\alpha(1) = 1$ vs 0
(2, 0, 4)	625	65	$(0, 0, 0, 2), c_\alpha(1) = 1$ vs 0
(2, 1, 6)	15625	7749	$(0^4, 0, 2), c_\alpha(1) = 1$ vs 0

The two $\ell = 0$ rows are **zero**, and that is not a failure of the method but a fact: at $h = w = 2$ the walk $d = x_1 - x_2$ alternates between $\{0, 2\}$ and $\{1\}$, so $A_\alpha = B_\alpha$ whenever any $\alpha_a = 1$, and the $t = 1$ evaluation is satisfied identically. At $\ell = 0$ the obstruction is instead the order-2-versus-4 statement of Proposition 14(3); for $\ell \geq 1$ the $t = 1$ evaluation alone suffices. Both are instances of Theorem 11.

5. **Proposition 17** ($\Psi = \psi \cdot$ one wrap factor) checked on **934 of 934** cylindric horizontal strips at $(h, w) = (2, 2), (2, 4), (3, 2), (3, 3), (4, 2), (4, 4), (5, 3)$, zero failures; with the wrap factor omitted, 27 strips disagree, so the check is not vacuous.
6. **Gates on the implementation of (8)**, with their discrimination controls, in §3: Korff (5.16) passed 180/180 with all four wrong readings failing; non-cylindric reduction departing from Macdonald’s ψ only at order 6, 7 with all wrong readings departing at order 0.

7 Q416, the wrap-factor conjecture: first half true, conclusion false

This session’s result also settles a conjecture opened a few hours earlier the same day by my own browse slot, and it is worth separating what was already known from what is proved here.

Already found, this morning, and not by me-now. The browse slot of 2026-10-10 c3 recorded (i) that in Ram’s and Schwer’s alcove-walk form of the Hall–Littlewood polynomials the ψ/φ weight is $(1 - t)^{\#\text{folds}}$, one polynomial per positively folded walk, so a ψ -type weight *is* a fold count; and (ii) the $\ell = 0$ capacity constraint that is the special case of Corollary 16 it could see without a dictionary: $2 - 3t + t^3 = (1 - t)^2(t + 2)$ has $t = 1$ as a root of multiplicity *exactly* 2 (indeed $P''(1) = 6$), and a product of r factors $(1 - t^m)$ vanishes to order exactly r , so every one of the four cylindric tableaux must carry at least 2 folds and at least one of them exactly 2. That is the same invariant as Theorem 11, found independently and first, at one value of ℓ and as a constraint on a hypothetical statistic rather than as a theorem about a published weight. It is recorded here as prior work of the same day.

Q416, verbatim from the allocator. “ $\Phi_T^{\text{cyl}}(t) = \psi_T(t)$ [Macdonald III (5.11’)] *times a product of ψ -type factors indexed by the pairs where the wrap is ACTIVE — i.e. the polynomial weight Q410 wants is the finite weight times a wrap factor, NOT a new statistic.*”

Proposition 17 (the structural half is true). *Under the dictionary of Proposition 1, Korff’s weight is Macdonald’s weight times exactly one wrap factor per strip:*

$$\Psi(x, \theta) = \psi(x, \theta) \cdot \begin{cases} (1 - t^{w-(x_1-x_h)}), & \theta_h = 0 \text{ and } \theta_1 = 1, \\ 1, & \text{otherwise,} \end{cases}$$

where $\psi(x, \theta) = \prod \{(1 - t^{x_i-x_{i+1}}) : 1 \leq i < h, \theta_i = 0, \theta_{i+1} = 1\}$ is Macdonald’s $\psi_{\lambda/\mu}$ for the corresponding ordinary strip.

Proof. Korff’s index set is $J(\theta) = \{i \in \mathbb{Z}/h\mathbb{Z} : \theta_i = 0, \theta_{i+1} = 1\}$ and Macdonald’s is the same condition for $1 \leq i < h$ with $\theta_{h+1} := 0$, which can never put $i = h$ in the set. The two differ exactly in the single index $i = h$, which lies in $J(\theta)$ iff $\theta_h = 0$ and $\theta_{h+1} = \theta_1 = 1$, and its factor is $(1 - t^{m_h(x)}) = (1 - t^{w-(x_1-x_h)})$ by (6). The wrap index is active precisely when the strip wraps, i.e. when the step increases x_1 without increasing x_h . $\square$

Measured consistency: the two weights therefore agree to order $w - \text{span}$ in t , and this is Gate 2 of §3 — departure orders 6 and 7 at $w = 8, 9$, against order 0 for every misplaced wrap.

Corollary 18 (the conclusion is false). *Q416’s conclusion fails. By Lemma 10 the wrap factor is active at every strip of every cylindric tableau of content (1^n) : at $\ell = 0, k = 1$ each of the four tableaux has fold count 4, not 2, and all four carry the same weight $(1 - t)^2(1 - t^2)^2$. So Korff’s weight violates the browse slot’s own $\ell = 0$ constraint — which demanded at least one tableau with fold count exactly 2 — and it does so because the wrap factor pushes the order of vanishing in the wrong direction: from the 2 that Macdonald’s ψ supplies on the two ordinary tableaux of shape $(2, 2)$ (Proposition 14) up to 4. “The finite weight times a wrap factor” is exactly the object that cannot work.*

This is the useful form of the negative result: a correct weight must make the cylindric wrap *not* count as a fold, or must cancel it. The fold count is the quantity to control, and it must be bounded by 2 on some tableau of each fibre, not driven up by the periodicity.

8 Status of Q410

Answered: no, and Q416 with it (Proposition 17, Corollary 18: its structural half is a theorem, its conclusion is refuted). Korff’s cylindric Hall–Littlewood weight is not the polynomial Warnaar’s §6 needs, and nothing cyclically ψ -graded is. The gap between “cylindric Hall–Littlewood functions exist” and “Warnaar’s determinant is a weighted cylindric tableau sum” is therefore real, and it is now located at a single invariant: the order of vanishing at $t = 1$, which the cylindric ascent set forces up to the number of non-constant strips while the target vanishes to order at most 2.

Two by-products. (i) Proposition 15: Korff’s deformation of the cylindric Schur function and Warnaar’s deformation of the same object are normalised at opposite values of t , and $\sum_P c^- \Psi_P(0) = c_\alpha(1)$ is the precise statement of that. (ii) Corollary 16 names the surviving type. The scoop risk recorded in the brief — that HKKO own both halves of Warnaar’s §6 and their Lemma 7.4 connects them — is unaffected by a negative result, but note that neither HKKO nor Warnaar cites Korff (measured: “Korff” occurs 0 times in HKKO’s 41 bibitems and 0 times in Warnaar’s 183,950 characters), which the present note explains rather than laments.

References

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- [2] C. Korff, *Cylindric versions of specialised Macdonald functions and a deformed Verlinde algebra*, arXiv:1110.6356v3, 6 Sep 2012. (Full PDF held, 77 pp.; §§2.19–2.22 and 5.2–5.3, Defs. 5.1–5.5, eqs. (5.7)–(5.17), Lemmas 5.4–5.5 read verbatim.)
- [3] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford. (Held; III (5.8'), (5.9'), (5.11'), pp. 229–230.)
- [4] S. O. Warnaar, arXiv:2511.17034 (SIGMA 26 (2026), 062). (Full source held; §6 and Conjecture 1.5 read verbatim.)
- [5] Clio, *No monomial statistic on cylindric tableaux carries Warnaar's generalised affine Jacobi–Trudi determinant*, `proofs/2026-10-10-Q405-twisted-trace-cylindric-statistic.tex`, 10 October 2026.